2017 EXAMINATIONS

Lancaster University

Part II

COMPUTING AND COMMUNICATIONS [2.5 Hours]

SCC.341 Game Design

Candidates are asked to answer THREE questions from FOUR; each question is worth a total of 25 marks.

Use a separate answer book for each question.

Question 1

1.a Callois' categorization of play consists of Agon, Alea, Mimesis and Ilinx. Using one example game for each category, briefly explain how the central game-play element relates to the category. Also suggest a modification to the game such that it also supports one more category of play. The game examples should include one of each of the following types: Board-based games, Card games, Video games, Playground games.

[10 marks]

1.b Consider David Parlett's games rules analysis diagram below:

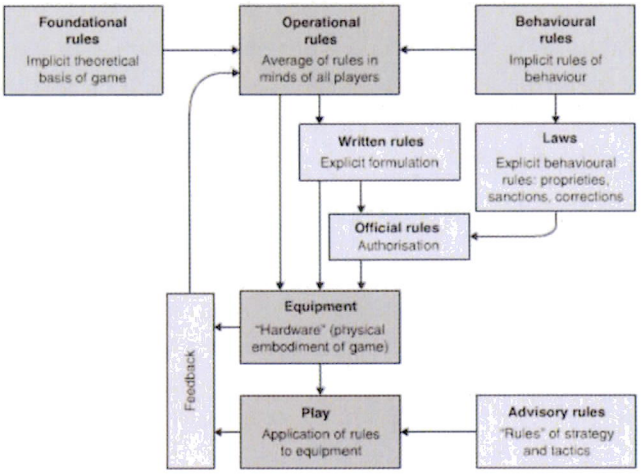

Figure 1.b: David Parlett's rules analysis

i. Describe the process involved in the figure above with particular focus on operational (stated), foundational (constitutive), behavioural (implicit) rules.

[3 marks]

ii. Choose a game based on a system of information and uncertainty. Identify its rules on three levels: operational, foundational and behavioural rules.

[3 marks]

iii. Explain how the game manipulates information where this information can be randomized, hidden, revealed, acquired and transformed.

[5 marks]

iv. Discuss how uncertainty and information theory are linked in the game.

[4 marks]

Total 25 marks

Question 2

2.a Which element of the Schell's game tetrad does the "Hero's Journey" fit into? Using an example of a RPG video game discuss how the tetrad can be satisfied without using "Hero's Journey".

[5 marks]

2.b Which two elements of the game tetrad are associated with the use of Game Physics? Discuss using examples how improper implementation of game physics affects these tetrad elements.

[5 marks]

2.c Using video game examples, describe where AABB, SAT and GJK algorithms for collision detection are suitable for use. Also contrast the pros and cons of using these algorithms.

[5 marks]

2.d Provide the definition of meaningful play in descriptive and evaluative contexts.

[2 marks]

2.e From a design viewpoint, games need to consider feedback mechanisms:

i. Identify how feedback as a system is represented in a game's state structure.

[2 marks]

ii. Using a game as an example, examine how the design rules formulated by Marc LeBlanc relating to feedback mechanisms can be used to limit or enhance the evaluative definition of meaningful play.

[6 marks]

Total 25 marks

Question 3

3.a List the different types of gamers described in Bartle's Gamer Personas and Lazarro's Gamification User Types.

[4 marks]

3.b. Compare and identify similarities between Bartle's Gamer Personas with the 'Girl Gamer' personas.

[6 marks]

3.c Using the example of an existing game, discuss how the game offers one of Le Blanc's categories of pleasure. Suggest a modification to the game such that the game can offer a previously unsupported category of pleasure.

[5 marks]

3.d Consider the example of the 'Prisoner's dilemma' and its optimal solution where each player maximizes their self-interest.

i. Discuss how the principles of game theory games are manifested in the 'Prisoner's dilemma'.

[4 marks]

ii. Create a decision matrix based on 'Prisoner's dilemma' labelling the player outcomes and deducing their utility values.

[2 marks]

iii. Assign payoffs to these utility values and use these to label a pay-off matrix.

[4 marks]

Total 25 marks

Question 4

4.a. You are tasked with designing a smartphone based game. The central gameplay element should involve Agon between players. Discuss the design of this gameplay element using the Ludic funnel and show how it satisfies two of Shigeru Miyamoto's design principles.

[6 marks]

4.b. Since the gameplay element involves two players using their own smartphones, list two situations where either player may see the other player disappear. For each of these situations, describe your design approach for handling both instances of the game.

[4 marks]

4.c Studying Figure 4.c below, explain the concept of complexity and explain how it differs from fixed, periodic and chaotic systems.

[4 marks]

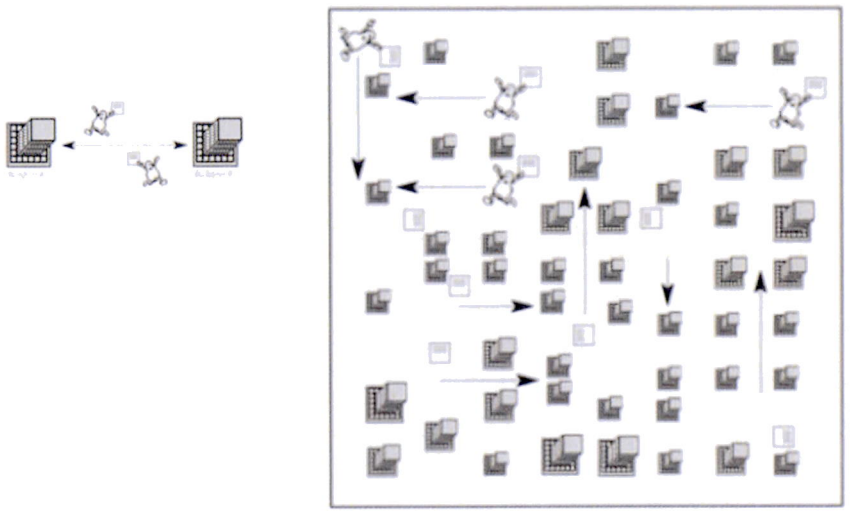

Figure 4.c: Examples of messengers delivering messages between buildings.

i. Choose an existing simple game and propose how your chosen game can become more complex by changing its rules by introducing three additional variants.

[6 marks]

ii. Analyse and explain how the three variants attain meaningful play.

[3 marks]

iii. Identify and discuss how your game demonstrates a degree of emergence.

[2 marks]

Total 25 marks

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